

Amitesh Puri

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West Bengal, India · 721433

Education

Generative Modelling from First Principles <i>Ranked 6th in the final project assessment</i>	IISc 2026
Mathematical Foundations of Machine Learning <i>Consolidated score: 68%</i>	NPTEL · IISc Bangalore Jan – Apr 2026
Manipal Academy of Higher Education (MIT Manipal) <i>M.Sc. in Applied Mathematics and Computing CGPA: 7.28 / 10</i>	Udupi, Karnataka, India Aug 2023 – Jun 2025
Ramnagar College, Vidyasagar University <i>B.Sc. in Mathematics Final Grade: 62%</i>	Paschim Medinipur, West Bengal, India Jul 2017 – Oct 2020

Research Experience

Physics-residual Flow Matching under distribution shift [\[Report\]](#) [\[GitHub\]](#)
Independent research report

- Ran a controlled comparison of a plain flow-matching baseline, a weighted physics residual (RFM), and the conflict-free PBFM formulation on a next-step generative task, across 5 seeds, 7 sampling budgets, and 4 graded distribution-shift families, with 95% bootstrap confidence intervals.
- In distribution, the one-line weighted residual ($\lambda = 0.005$) lowered relative L2 from 0.0148 to 0.0094 (about 37%) and the PDE residual from 13.2 to 7.5 (about 44%), with MMD down about 63%, at a matched 16-step sampling budget.
- Found the parameter-free PBFM statistically indistinguishable from the baseline, and reported that the residual's advantage did not carry over under stronger initial-condition and forcing shifts, where it often reversed.

When does a geodesic probability path help latent flow matching? [\[Report\]](#) [\[GitHub\]](#)
Independent research report

- Tested whether a straight-line or a geodesic probability path changes latent flow matching, holding the data, model, optimiser, and seed fixed across three converged, seed-matched runs.
- Reached a controlled null result: the two paths were statistically indistinguishable on accuracy and, once seed noise was removed, on the spectrum, and an early single-seed lead for the straight line did not survive repetition.
- Isolated the cause with an ablation: geodesic geometry helped only on manifold-structured data (geodesic penalty 0.63) and hurt off it (2.17), and measured the real VQ-VAE latents to be off the sphere (radius CV ≈ 0.27 , intrinsic dimension ≈ 5.5 against an ambient 32).

Master's Research

Addressing Conditional Shift in Graph Data: An Empirical Study of [\[Report\]](#) [\[GitHub\]](#)

Distance Metrics

Master's Research Report, Manipal Academy of Higher Education Supervisor: Dr. Akansha Singh

- Studied the robustness of semi-supervised node classification under conditional distributional shift across several GNN architectures on benchmark datasets.
- Added distance metrics as auxiliary regularisers that penalise divergence between training and test node representations, isolating each metric's effect in controlled ablations.
- Used conformal prediction to evaluate coverage guarantees under distributional change.
- Implemented all experiments in PyTorch with fixed random seeds and publicly available code.

Technical Training

Guided Projects Student

Nov 2022 – Feb 2023

The Machine Learning Company

Remote

- Completed applied projects in NLP and computer vision, including end-to-end pipelines for text classification, sentiment analysis, and language generation.
- Trained and evaluated deep learning models for image classification and object detection tasks.

Data Science Trainee

Aug 2021 – Jan 2022

Evarcity

Bangalore, India

- Built Python-based data analysis and machine learning workflows on real-world datasets, with practical exposure to feature engineering, model evaluation, and data visualisation.

Personal Projects

Cyberbullying Detection

[\[GitHub\]](#)

- Developed a text classifier in TensorFlow for identifying cyberbullying in social media content. Deployed the model through a Gradio interface, with user feedback collected and managed using Argilla for iterative data annotation.

Text Generation with GPT-2

[\[GitHub\]](#)

- Fine-tuned GPT-2 on a domain-specific corpus and deployed an interactive text generation interface for user evaluation and qualitative assessment of output quality.

English to Hindi Machine Translation

[\[GitHub\]](#)

- Fine-tuned a transformer-based sequence-to-sequence model for English to Hindi translation and deployed the system through a Streamlit application with integrated text-to-speech output.

Technical Skills

Programming	Python, R
ML / DL Frameworks	PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, PyCaret
Data Analysis	Pandas, NumPy, Matplotlib, Seaborn, Data Visualization
Tools	Git, Gradio, Streamlit, Hugging Face Hub, Argilla